INISIATIF KESIHATAN PARU-PARU KEBANGSAAN (LHI) · ALIRAN PESAKIT DARI KLINIK KE HOSPITAL

# Integrated Approach to Lung Health Screening and Management

KEMENTERIAN KESIHATAN MALAYSIA
JABATAN KESIHATAN NEGERI KEDAH

## PRIMARY CARE · KLINIK KESIHATAN

**RED FLAGS**
- Unexplained chronic cough
- Stridor
- Unexplained weight loss
- Worsening breathlessness
- Recurrent chest infection
- Frank haemoptysis

Start → Risk Factors for Lung Disease → Symptoms? (Yes → Red flags? (Yes → Refer specialist (FMS / Internal Medicine / Respiratory Physician)); No → Red flags? (No)) → Detailed history, examination and risk assessment

Detailed history, examination and risk assessment → Risk for Asthma / Risk for COPD / Risk for Lung Cancer / Risk for Tuberculosis

**Risk for Asthma**
- Spirometry
- Normal? (Yes → Clinically suspected asthma? (No → Follow up as indicated; Yes → Elevated biomarkers*? (No → Refer specialist for further evaluation; Yes → Asthma / COPD)); No → Obstructive → Asthma / COPD)

**Risk for COPD**
- PUMA (≥5) → Spirometry
- Normal? (No → Obstructive → Asthma / COPD; Restructive → Refer specialist for further evaluation)

Asthma / COPD → Treatment and follow up

**Risk for Lung Cancer**
- CXR (with or without AI-assisted)
- Normal? (Yes → Low-dose CT → Lung-RADS 0, 1, 2, 3 → Surveillance; Lung-RADS 4A, 4B, 4X, S; No → Nodule / Mass; Consolidation, atelectasis, pleural effusion etc)

**Risk for Tuberculosis**
- Symptomatic → Follow CPG for TB work-up
- Asymptomatic → Refer: High Risk Group TB Screening Guidelines

## SPECIALIST CARE · HOSPITAL

Treatment and follow up → Difficult to treat / Frequent exacerbator

- Difficult to treat → Referral via Severe Asthma Network → MDT → Decision for biologic → Follow up
- Frequent exacerbator → Refer to specialist → MDT → Optimise treatment · Decision for biologic → Follow up

Lung-RADS 4A, 4B, 4X, S / Nodule / Mass / Consolidation, atelectasis, pleural effusion etc → Refer specialist for further evaluation

- CECT / PET scan → Confirmatory investigation → Basic & advanced bronchoscopic procedure / image-guided biopsy / pleuroscopy / surgical resection → Confirmed histopathology → Molecular testing → MDT → Decision for treatment

Other lung diseases → Investigate and treat as indicated

***Biomarkers:** blood eosinophil count, total and specific IgE, FeNO (fraction of exhaled nitric oxide).

**PUMA:** COPD case-finding questionnaire; score ≥5 → spirometry.

**MDT:** multidisciplinary team.

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